Approximating Irrational Roots on a Number Line
Answer Key

Grade 8 / Pre-Algebra The Number System

Quick Answers

1. 6, 7
2. 12, 13
3. 5.5
4. —
5. 9.1
6. 7.6
7. 2.7, $2\sqrt{2}$, 3.1, $\sqrt{10}$
8. 6.4, 6.7
9. 11.5
10. 7.1

Curriculum

Level: Grade 8 / Pre-Algebra The Number System
8.EE.A – *problem 9*
8.NS.A – *problems 1, 2, 3, 4, 5, 6, 7, 8, 10*
Difficulty 1–5 of 5 across 13 tagged checks.

Common wrong answers

6. *If they got 7.4: picked the root of the smaller radicand, which lands a full two tenths to the left*
6. *If they got 7.9: picked the root of the larger radicand, which lands to the right of the marked point*
10. *If they got 7.0: rounded to the root of the nearer perfect square instead of estimating between the two brackets*

1. Between which whole numbers is $\sqrt{47}$?

Step 1. 47 is not a perfect square, so the root is irrational and has to be trapped between the two perfect squares on either side of it.
Step 2. $6^2 = 36$ and $7^2 = 49$. Since $36 < 47 < 49$, taking square roots (which preserves order for positive numbers) gives $6 < \sqrt{47} < 7$. $\boxed{6 < \sqrt{47} < 7}$
Step 3. On the line, 47 is much nearer 49 than 36, so the dot A belongs close to 7 — just left of it, around 6.9.

2. Between which whole numbers is $\sqrt{150}$?

Step 1. Same method, higher up the line: find the perfect squares that bracket 150.
Step 2. $12^2 = 144$ and $13^2 = 169$, and $144 < 150 < 169$, so $\boxed{12 < \sqrt{150} < 13}$
Step 3. 150 sits only a little above 144, so the dot B belongs just to the right of 12 — about a quarter of the way to 13.

3. $\sqrt{30}$ to the nearest tenth.

Step 1. Bracket first: $5^2 = 25$ and $6^2 = 36$, so the root is between 5 and 6. Now test tenths by squaring, since squaring is the only tool available without a calculator.
Step 2. $5.4^2 = 29.16$ (too small) and $5.5^2 = 30.25$ (too big), so $5.4 < \sqrt{30} < 5.5$.
Step 3. 30 is 0.84 above 29.16 but only 0.25 below 30.25, so the root is nearer the upper tenth: $\boxed{\sqrt{30} \approx 5.5}$

On the line: the dot goes just left of the 5.5 tick.

4. Compare $\sqrt{20}$ and 4.5.

Step 1. One number is irrational and one is a terminating decimal, so put them in the same form: square the decimal and compare with the radicand.
Step 2. $4.5^2 = 20.25$, and $20 < 20.25$. Taking square roots keeps the order, so $\sqrt{20} < 4.5$.
Step 3. $\boxed{\sqrt{20} < 4.5}$

On the line: $\sqrt{20} \approx 4.47$, so its dot sits between the 4.4 and 4.5 ticks — close to 4.5 but definitely to its left. Squaring the decimal, rather than estimating the root, is what makes the comparison exact.

5. $\sqrt{83}$ to the nearest tenth.

Step 1. $9^2 = 81$ and $10^2 = 100$, so $9 < \sqrt{83} < 10$, and 83 is close to the bottom of that gap.
Step 2. $9.1^2 = 82.81$ (too small) and $9.2^2 = 84.64$ (too big), so $9.1 < \sqrt{83} < 9.2$.
Step 3. 83 is only 0.19 above 82.81 but 1.64 below 84.64, so the root rounds down:

$$\boxed{\sqrt{83} \approx 9.1}$$

On the line: the dot sits just right of the 9.1 tick.

6. Which of $\sqrt{55}$, $\sqrt{58}$, $\sqrt{62}$ belongs at P?

Step 1. P is the sixth tick past 7.0, so P is at 7.6. Rather than estimate three roots, square the position: whichever radicand is closest to 7.6^2 is the one that belongs there.
Step 2. $7.6^2 = 57.76$ and $7.7^2 = 59.29$. Only 58 lies between those, so $\sqrt{58}$ is the root whose value rounds to 7.6.
Step 3. $\boxed{\sqrt{58} \approx 7.6}$

Common wrong answers: 7.4 — $\sqrt{55}$, which lands two ticks left of P; and 7.9 — $\sqrt{62}$, which lands to the right of P. Squaring the tick value tests all three candidates at once.

7. Order 2.7, $\sqrt{8}$, 3.1, $\sqrt{10}$ from least to greatest.

Step 1. Mixed rationals and irrationals cannot be compared by appearance, so estimate each root to a tenth first and read the order off the line.
Step 2. $2.8^2 = 7.84$ and $2.9^2 = 8.41$, so $\sqrt{8} \approx 2.8$. And $3.1^2 = 9.61$ while $3.2^2 = 10.24$, so $\sqrt{10} \approx 3.2$.
Step 3. Placing 2.7, 2.8, 3.1 and 3.2 on the line in that order gives

$$\boxed{2.7 < \sqrt{8} < 3.1 < \sqrt{10}}$$

Watch the near miss: 3.1 and $\sqrt{10}$ look close, but $3.1^2 = 9.61 < 10$, so $\sqrt{10}$ is genuinely the larger of the two.

8. Plot $\sqrt{41}$ and $\sqrt{45}$ to the nearest tenth.

Step 1. Both radicands lie between 36 and 49, so both roots lie between 6 and 7 — the whole number line printed for this problem.

Step 2. $6.4^2 = 40.96$ and $6.5^2 = 42.25$, so 41 is barely above 40.96: $\boxed{\sqrt{41} \approx 6.4}$

Step 3. $6.7^2 = 44.89$ and $6.8^2 = 46.24$, so 45 is barely above 44.89: $\boxed{\sqrt{45} \approx 6.7}$

On the line: both dots sit just to the right of their ticks, which is the visual signature of a radicand only slightly above a tenth's square.

9. Square patio of area 132 square feet — side to the nearest tenth of a foot.

Step 1. A square's side is the square root of its area, so the side length in feet is $\sqrt{132}$, and 132 is not a perfect square.

Step 2. $11^2 = 121$ and $12^2 = 144$, so the side is between 11 and 12 feet. Test tenths: $11.4^2 = 129.96$ (too small) and $11.5^2 = 132.25$ (too big).

Step 3. 132 is 2.04 above 129.96 but only 0.25 below 132.25, so the side rounds up:

$\boxed{\sqrt{132} \approx 11.5 \text{ ft}}$

Sense check: 11.5 feet squared is about 132.25 square feet, which is within a quarter of a square foot of the real patio — a sensible carpentry-grade estimate.

10. Why Nadia's rounding rule fails for $\sqrt{50}$.

Step 1. Nadia rounded the *radicand* to the nearest perfect square and then took its root. That is not the same operation as rounding the root, and the two disagree whenever the radicand sits near the low end of its bracket — which is exactly this case.

Step 2. Bracket properly: $7^2 = 49$ and $8^2 = 64$, so $7 < \sqrt{50} < 8$. Now test tenths: $7.0^2 = 49$ (too small) and $7.1^2 = 50.41$ (too big), so $7.0 < \sqrt{50} < 7.1$.

Step 3. 50 is a full 1 above 49 but only 0.41 below 50.41, so the root is nearer the upper tenth: $\boxed{\sqrt{50} \approx 7.1}$

Step 4. The explanation: squaring stretches the gaps, so a radicand that is only a little above a perfect square already has a root a noticeable distance above that square's root. Nadia's rule throws away the size of the gap, and the number line shows $\sqrt{50}$ sitting clearly right of the 7.0 tick.

Common wrong answer: 7.0 — the root of the nearer perfect square, reported instead of an estimate between the brackets.